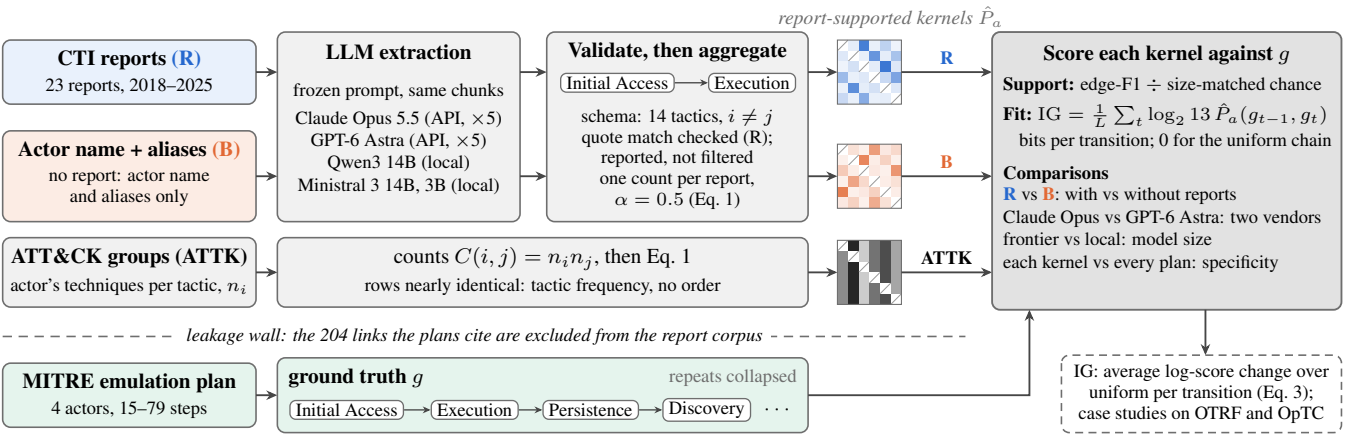

report-supported kernels $\hat{P}_a$
CTI reports (R)
23 reports, 2018–2025
Actor name + aliases (B)
no report: actor name
and aliases only
ATT&CK groups (ATTK)
actor's techniques per tactic, n_i
LLM extraction
frozen prompt, same chunks
Claude Opus 5.5 (API, ×5)
GPT-6 Astra (API, ×5)
Qwen3 14B (local)
Ministral 3 14B, 3B (local)
counts $C(i,j) = n_i n_j$, then Eq. 1
rows nearly identical: tactic frequency, no order
Validate, then aggregate
Initial Access → Execution
schema: 14 tactics, $i \neq j$
quote match checked (R);
reported, not filtered
one count per report,
$\alpha = 0.5$ (Eq. 1)
R
B
ATTK
Score each kernel against g
Support: edge-F1 ÷ size-matched chance
Fit: IG $= \frac{1}{L}\sum_t \log_2 13\, \hat{P}_a(g_{t-1}, g_t)$
bits per transition; 0 for the uniform chain
Comparisons
R vs B: with vs without reports
Claude Opus vs GPT-6 Astra: two vendors
frontier vs local: model size
each kernel vs every plan: specificity
leakage wall: the 204 links the plans cite are excluded from the report corpus
MITRE emulation plan
4 actors, 15–79 steps
ground truth g
repeats collapsed
Initial Access → Execution → Persistence → Discovery · · ·
IG: average log-score change over uniform per transition (Eq. 3);
case studies on OTRF and OpTC